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DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: ENGINEERING CAPSTONE PROJECT – 1(23IE4053R/23IE4053A)
A.Y. 2026 - 2027

TITLE OF THE PROJECT	INTELLIGENT SUPPLIER RISK PREDICTION AND ANALYTICS	
S. No.	**University ID**	**Name**
1	2320030092	GULLAPALLI SHEETAL
2	2320030098	JANGARI SAKSHITHA
3	2320030102	GULLAPALLI NIKITHA CHOWDARY
Name of the Guide		Dr N Shirisha

Abstract

The Intelligent Supplier Risk Prediction and Analytics project is developed to help organizations identify and understand supplier-related risks before they become major problems. The system uses supplier financial, operational, and news-related information to predict the overall risk level of suppliers and provide useful insights through an interactive dashboard. XGBoost is mainly used to predict financial and operational supplier risk, while FinBERT is used to analyze the sentiment of supplier-related news.

LSTM is used to study changes in sentiment over time and improve the risk assessment. The system also uses K-Means to group suppliers based on their performance, Isolation Forest to detect unusual supplier behavior, and SHAP to explain the main factors contributing to a supplier's risk.

The system supports two ways of working: users can upload their own supplier dataset in CSV or Excel format, or they can use a Live Data Mode that generates simulated supplier data for real-time risk monitoring.

The results are presented through a Streamlit dashboard with interactive charts, risk scores, supplier analysis, anomaly alerts, risk trends, and early-warning indicators.

The dashboard helps users easily identify high-risk suppliers and understand the reasons behind the predicted risk. Overall, the project provides a practical data-driven approach for monitoring supplier performance and supporting better supplier risk management decisions.

Signature of the Guide HOD